Where a population starts decides which peak it climbs

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Vellum example talk · illustrative

12 October 2026

Most random starts end on the lower of the two peaks

59 %

of 500 random starts end on the lower peak

0.72

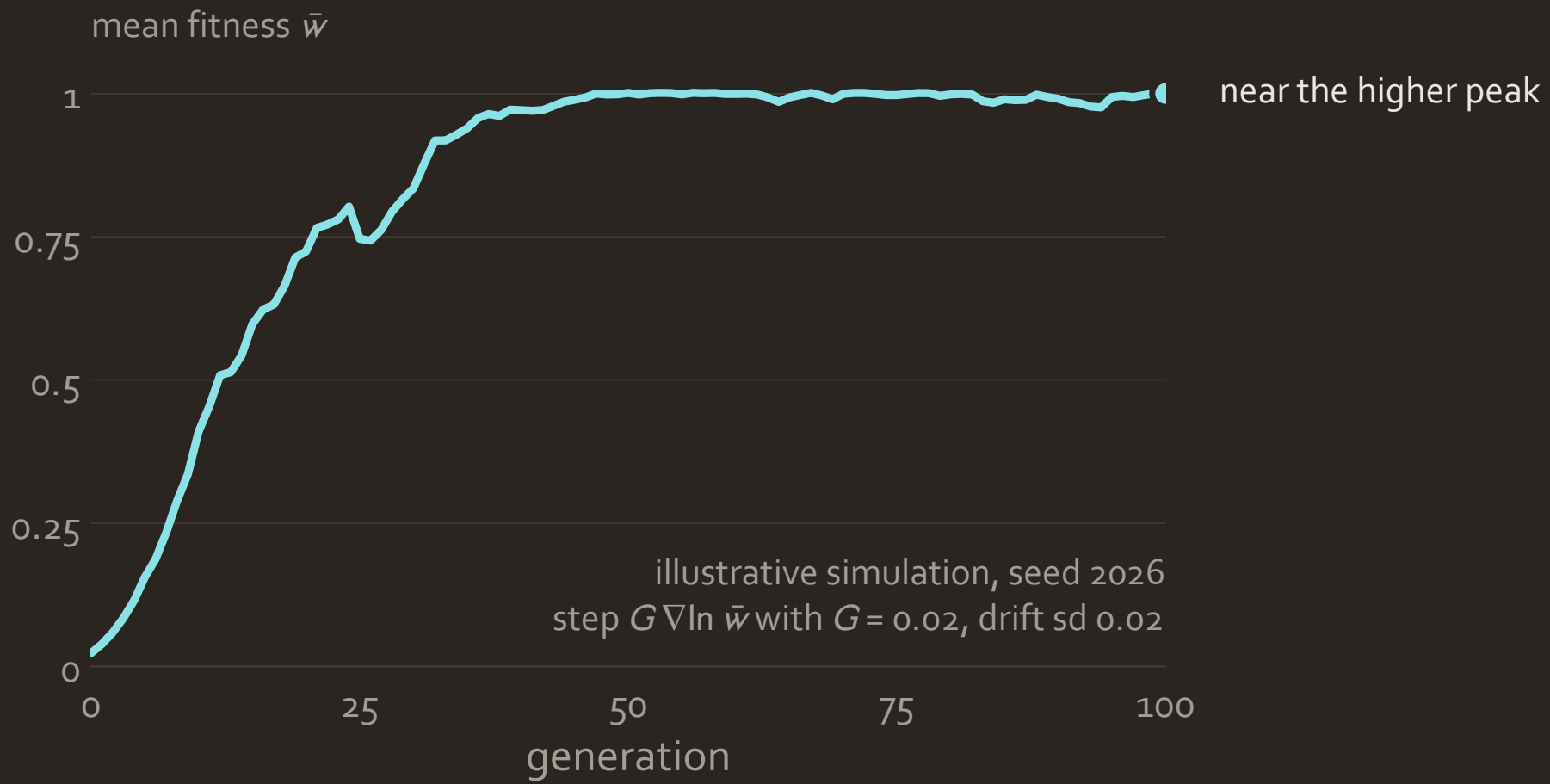
mean fitness at the top of the lower peak

1.00

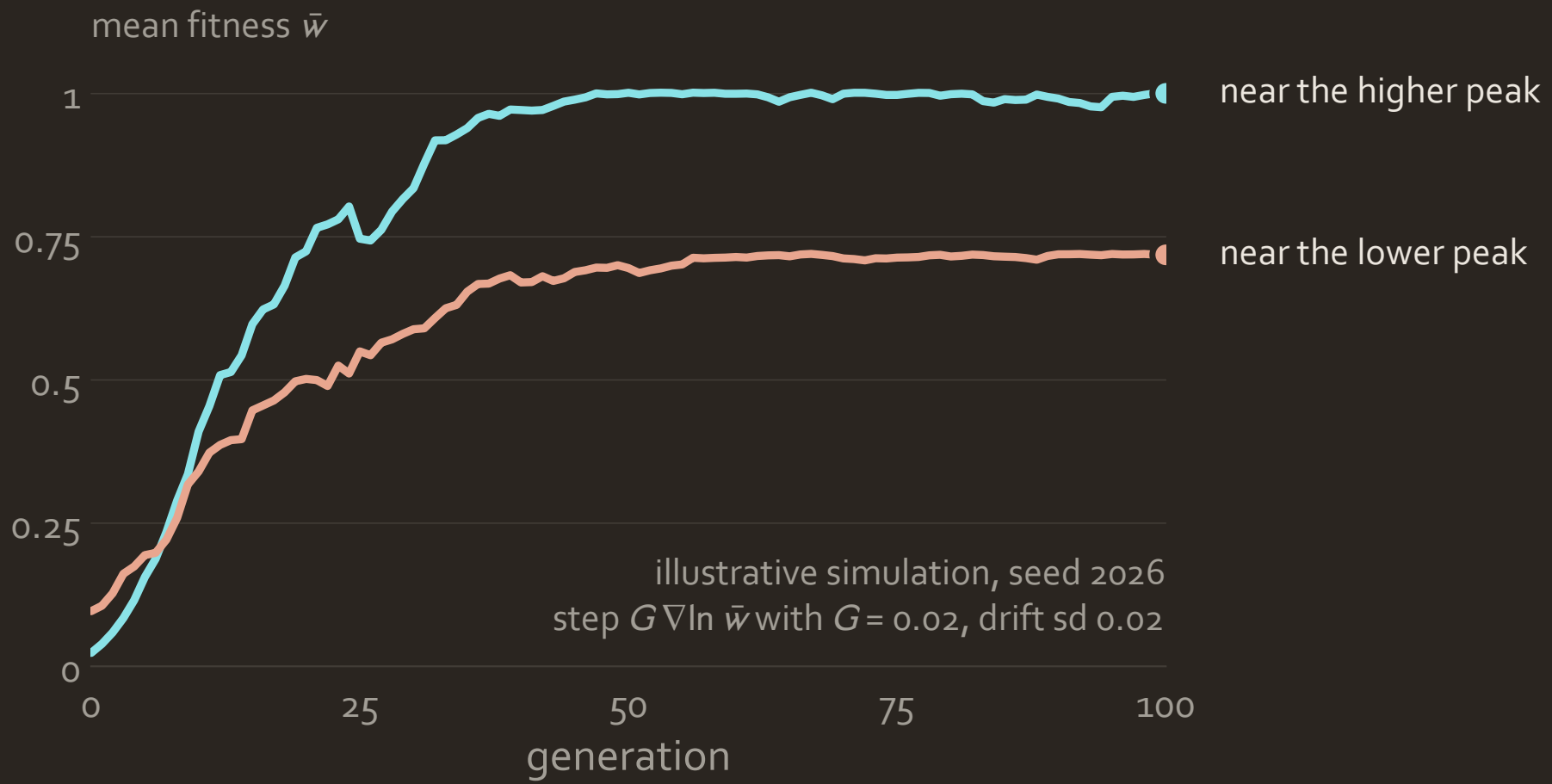
mean fitness at the top of the higher peak

Starts drawn evenly over the trait space, 100 generations each.

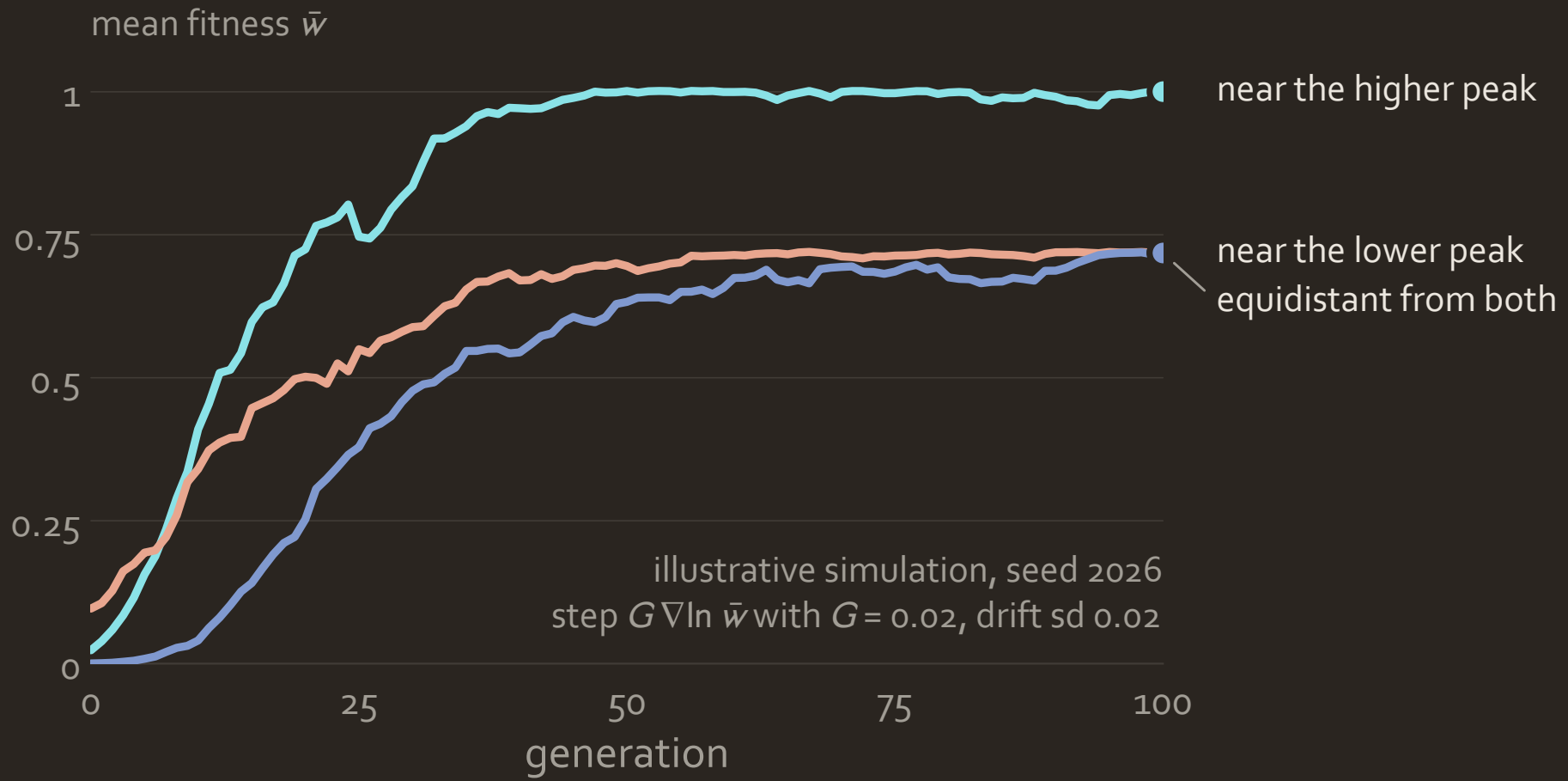
A start near the higher peak climbs it in 36 generations



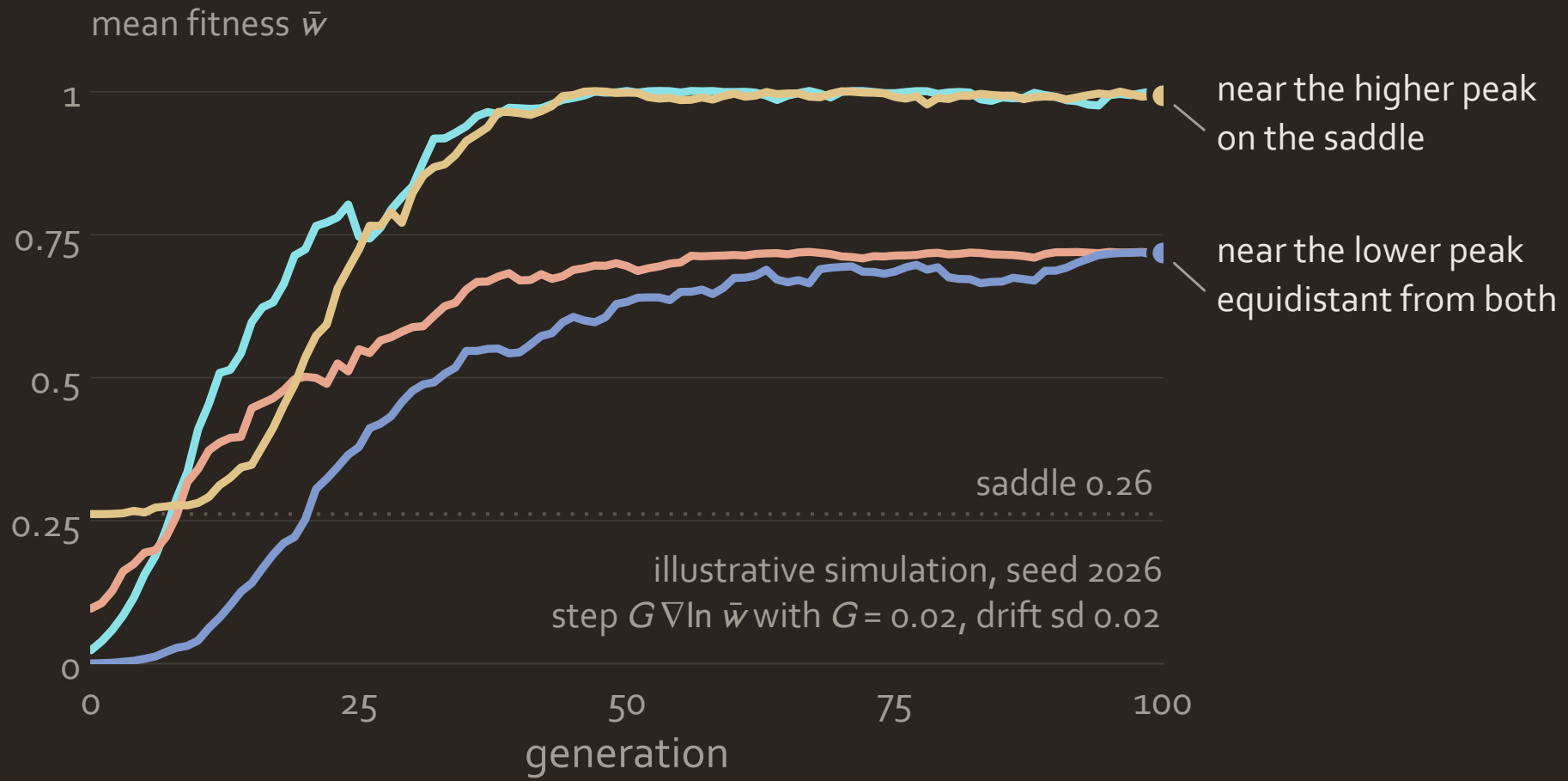
A start near the lower peak stops there, at 0.72



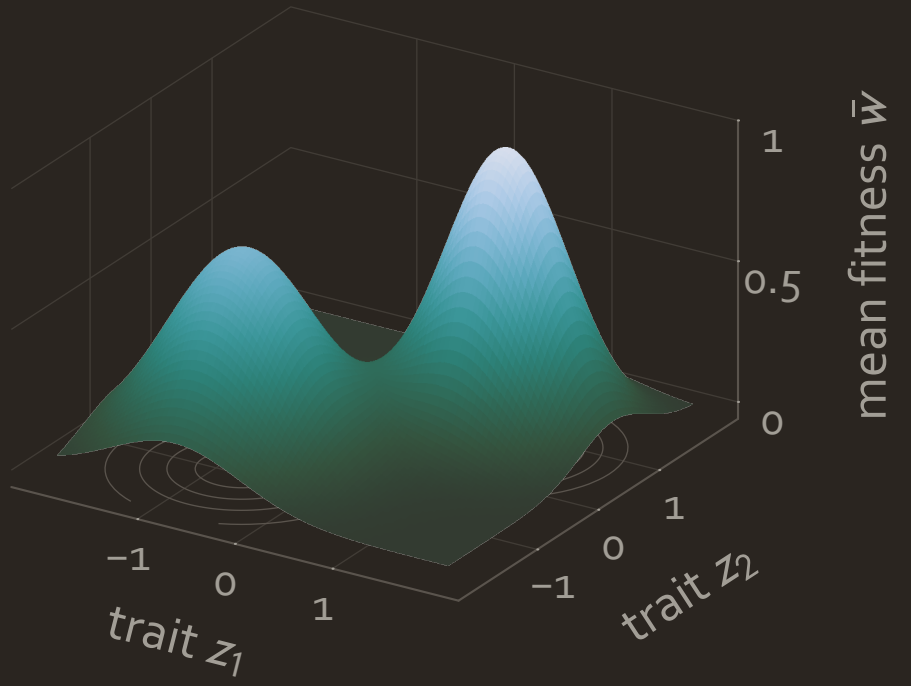
A start equidistant from both peaks ends on the lower one



A start on the saddle waits 12 generations, then climbs



The landscape has two peaks with a saddle between them



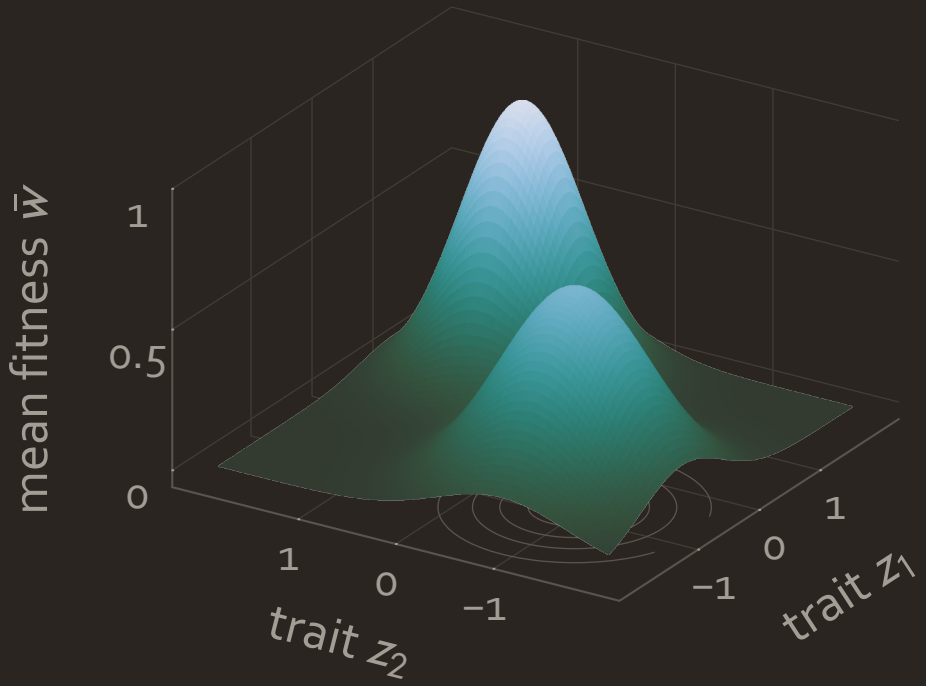
higher peak 1.00 at (0.90, 0.70)

lower peak 0.72 at (−0.80, −0.90)

saddle 0.26 at (0.05, −0.10)

elevation 24°, azimuth −58°

Turned by 90°, the higher peak stands 0.28 above the lower



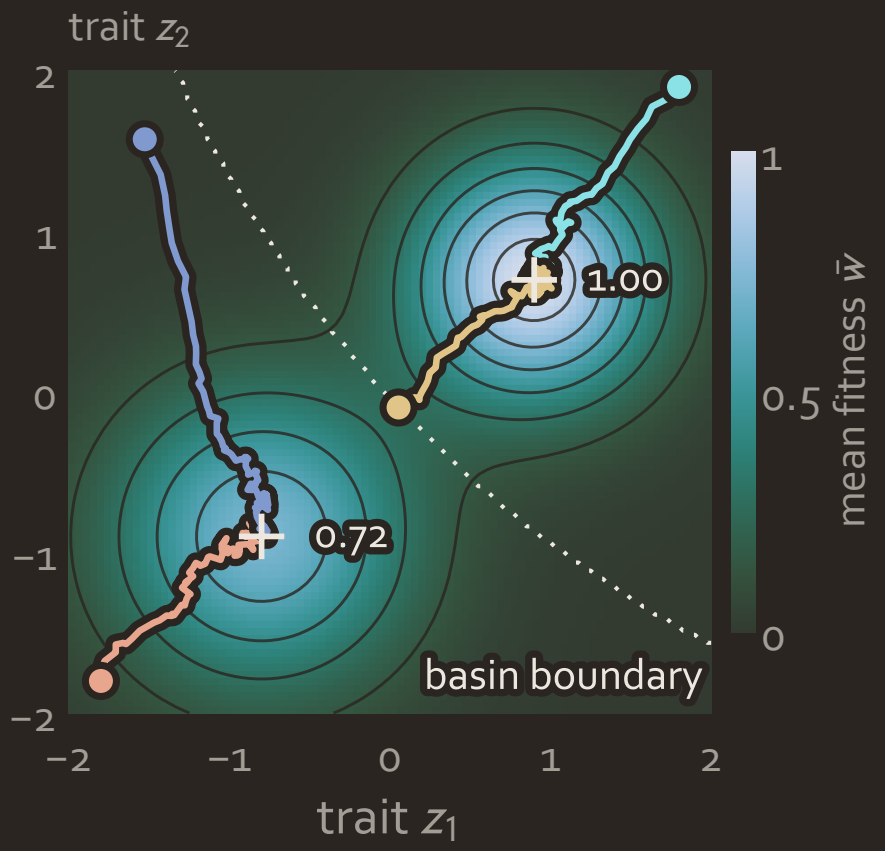
the lower peak now stands in front

the higher one rises behind it

the saddle hides between them

elevation 24°, azimuth −148°

Each population climbs the peak of the basin it starts in



ringed dot: where a population starts

cross: a peak and its height

dotted: the basin boundary

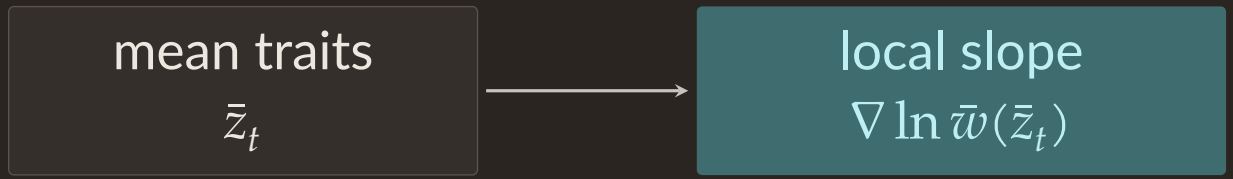
lower basin: 54 % of the map

Each generation moves the mean traits up the local slope

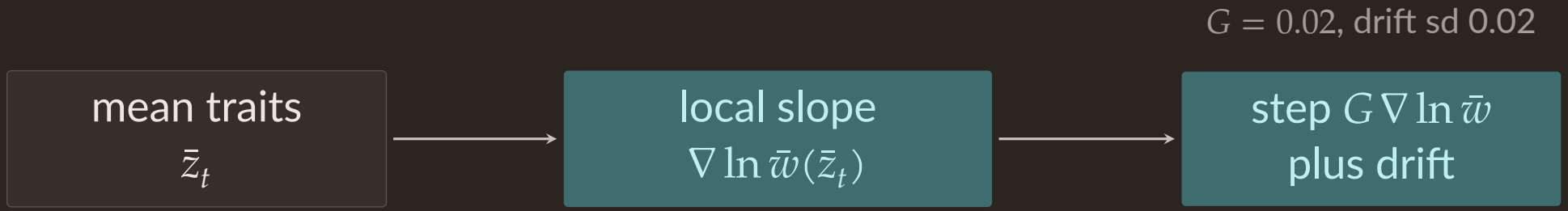
mean traits

$\bar{z}_t$

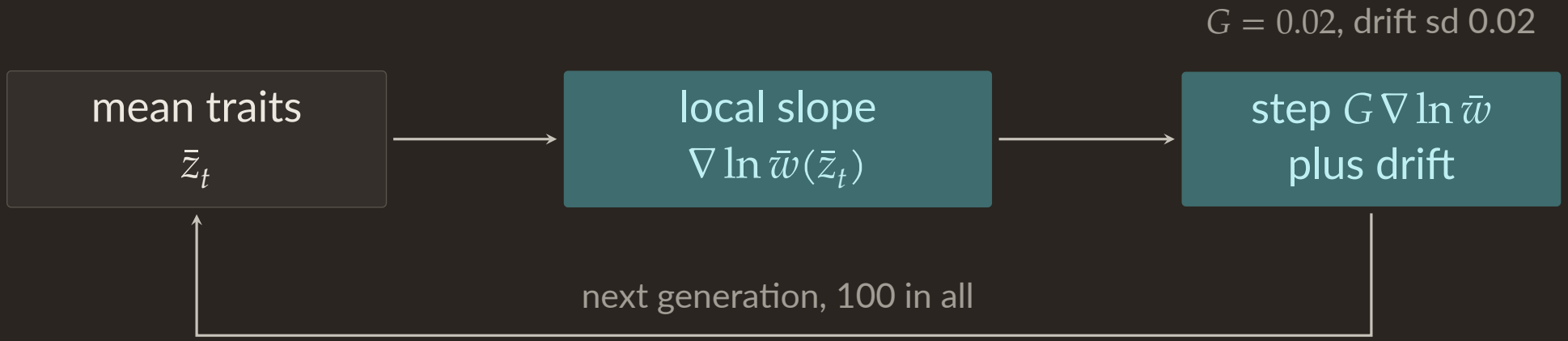
Each generation moves the mean traits up the local slope



Each generation moves the mean traits up the local slope



Each generation moves the mean traits up the local slope



Where the slope is flat, only drift moves the mean:
at a peak, or on the saddle.

The basin a population starts in decides its peak

The lower peak’s basin covers 54 % of the trait space; 59 % of random starts end on that peak, at 0.72, not on the higher one at 1.00.

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slides, figures and the simulation script are in the Vellum repository